FAE-SINDy: Discretization-Consistent Discovery of Latent Dynamics from Functional Data

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Motivation

Spatiotemporal systems are often measured as high-dimensional fields $u(x,t)$, yet many are governed by a few latent variables obeying simple equations. SINDy (Brunton et al.) recovers such equations from low-dimensional state measurements, SINDy-Autoencoder (Champion et al.) learns the latent coordinates jointly from a high-dimensional input.

Real fields are measured off-lattice: scattered, drifting, or failing sensors, and non-uniform or adaptively-refined meshes. There is no more reason to assume the right grid than the right variables.

We aim to recover a sparse, interpretable model of a dynamical system from functional data, independent of how the input is sampled.

Method

Functional Autoencoders

The encoder is a functional:

$$f(u;\theta) = \rho\left(\int_{\Omega} \kappa(x, u(x); \theta)\, dx; \theta\right) \in \mathbb{R}^{d_z}$$

The kernel κ lifts pointwise values, integration over Ω pools them into fixed-sized representation, ρ, maps to a low-dimensional latent representation. The pooling is approximated by a quadrature on any grid, so the encoder can be trained on one sampling and evaluated on another.

The decoder $g(z;\psi)$ reconstructs the field enforcing meaningful representations in the latent space.

SINDy

Given latent states $z(t)$ and their derivatives $\dot{z}(t)$, SINDy fits:

$$\dot{z}(t) = \Theta(z^T)\Xi$$

where Θ is a library of candidate terms and Ξ is a learned coefficient matrix. The ℓ_1 regularization term in the loss function (Figure 1b) in addition to sequential thresholding (STLSQ) drives Ξ sparse, so the model is parsimonious.

Contribution

We combine a functional encoder with SINDy, giving a discovery pipeline that is:

- **Discretization-consistent:** the encoder pools samples by quadrature, so the latent code — and hence the discovered dynamics - is independent of the sampling mesh up to quadrature error.
- **Interpretable:** the Champion four-term loss (ℓ_1 + STLSQ) keeps the latent ODEs sparse.

Preliminary Results

Lorenz-63 is lifted to a spatial field via six Legendre modes on N points. Evaluated at $2N$ with no retraining, the discovered latent model is unchanged and reconstruction error is 3.5×10^{-7}, evidence of mesh-transfer. The correct sparse structure is recovered up to one false negative ($-z_2$) and one false positive (z_1^2); surviving coefficients match the true dynamics to $< 1\%$. Both artifacts trace to STLSQ, which thresholds terms prematurely with no reactivation.

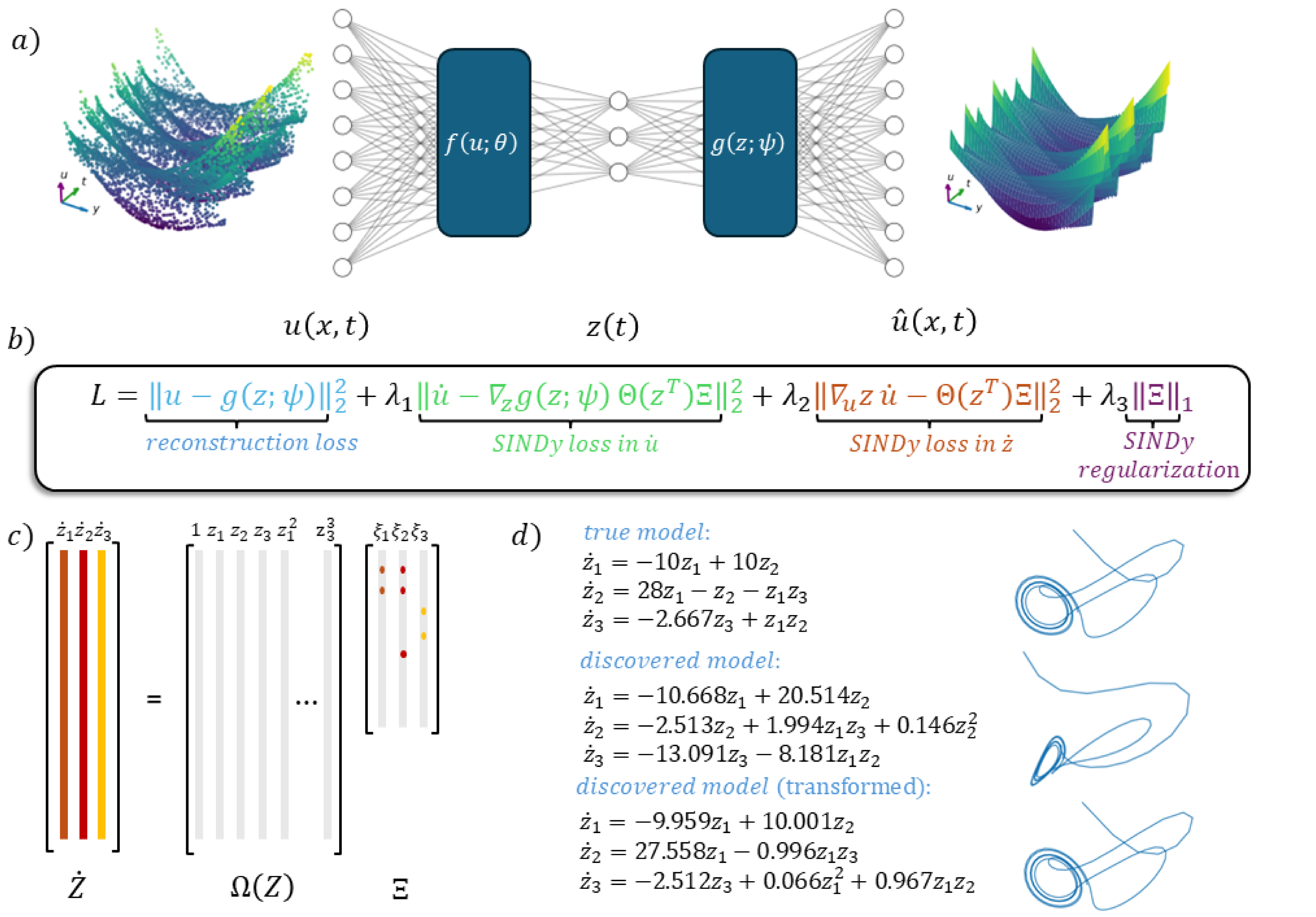

Figure 1: a) FAE-SINDy architecture: subsampled spatial samples $u(x)$ are mapped to the intrinsic coordinates $z(x)$. b) The pointwise loss function minimizes both the reconstruction error and the SINDy loss. c) A SINDy model learns a sparse representation of the latent dynamics up to a diffeomorphism. d) Comparison of true, recovered and transformed model and their visualization.

Applications

Discretization-consistency targets regimes where a fixed grid is unrealistic:

- **Multi-resolution data:** pooling snapshots across meshes or adaptively-refined simulations without re-gridding.
- **Model transfer:** inpainting, zero-shot super resolution, training acceleration
- **Reduced-order modelling of PDEs:** a mesh-independent latent ODE for spatiotemporal systems.

Conclusion and future work

FAE-SINDy recovers interpretable latent dynamics from functional data without grid dependence and allows.

- PDE experiments: Reaction-Diffusion, KS
- Architecture improvements SR3 ↔ STLSQ
- Quadrature error quantification

Key references

1. Brunton et al. (2016). Discovering governing equations from data by sparse identification of nonlinear dynamical systems. PNAS, 113(15), 3932–3937.
2. Bunker et al. (2025). Autoencoders in Function Space. JMLR, 26(165), 1–54.
3. Champion et al. (2019). Data-driven discovery of coordinates and governing equations. PNAS, 116(45), 22445–22451.

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